Equations & Inequalities: Mini-Lessons

NC.8.EE.7

ANSWER KEY

3.1 One- and two-step equations NC.8.EE.7

To **solve** an equation, get **x alone**. Undo each move with its **opposite**, and do it to **both sides** so the equation stays balanced.

Variable

a letter, like x, that stands for a number

Inverse (opposite) operation

the move that undoes another: + and − undo each other, × and ÷ undo each other

Solution

the number that makes the equation true

To undo...	+ 5	− 4	× 3 (3x)	÷ 2
do this	− 5	+ 4	÷ 3	× 2

Order: undo + or − first, then × or ÷.

Watch out: Do the same move to **both sides**. A move on one side only breaks the balance.

x = 4

Check: 2(4) + 3 = 11 ✓



Walk through: `Solve 3x - 4 = 11`

1. Which move comes first?
   Add 4 to both sides → 3x = 15
2. x is multiplied by 3. What undoes that?
   Divide both sides by 3 → x = 5
3. Check: put 5 in for x. What is 3(5) − 4?
   11, so x = 5 is right → 3(5) − 4 = 15 − 4 = 11 ✓

Check your understanding

1 Score 3 NC.8.EE.7

Solve: **x + 7 = 15**

Answer: 8

1. $x + \cancel{7} = 15$ ↓ Subtract 7
   $\cancel{-7} \quad -7$
2. $x = 8$

2 Score 3 NC.8.EE.7

What is the **first** move to solve **4x + 5 = 21**?

- (A) Add 5 to both sides
- (B) Divide both sides by 4
- (C) **Subtract 5 from both sides**
- (D) Subtract 4 from both sides

3 Score 3 NC.8.EE.7

Solve: **2x + 3 = 11**

Answer: 4

1. $2x + \cancel{3} = 11$ ↓ Subtract 3
   $\cancel{-3} \quad -3$
2. $\frac{\cancel{2}x}{\cancel{2}} = \frac{8}{2}$ ↓ Divide by 2
3. $x = 4$

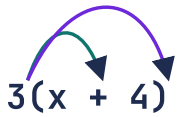
3.2 Distribute and combine like terms NC.8.EE.7

The number outside the parentheses **multiplies every term inside**. **Like terms** (x-terms with x-terms, numbers with numbers) can be added together.

Distribute
multiply the outside number by each term inside the ( )

Like terms
terms with the same variable part: 3x and 5x, or 4 and 9

Coefficient
the number in front of a variable (in 3x, the 3)



3 · x + 3 · 4

3x + 12

`4(x - 2) = 4x - 8`

`3x + 5 + 2x = 5x + 5`

Watch out: 3(x + 4) is **not** 3x + 4. The 3 multiplies the 4 too: 3x + 12.

Walk through: `Solve 2(x + 5) = 16`

1. There are parentheses. What is the first move?
   Distribute: 2 · x + 2 · 5 → `2x + 10 = 16`
2. Undo the + 10.
   Subtract 10 from both sides → `2x = 6`
3. Get x alone.
   Divide both sides by 2 → `x = 3`

Check your understanding

1 Score 3 NC.8.EE.7

Which is equal to `3(x + 4)`?

(A) 3x + 4
(B) x + 12
(C) **3x + 12**
(D) 3x + 7

2 Score 3 NC.8.EE.7

Combine like terms: `5x + 2 + 3x`

(A) $8x^2$
(B) 15x + 2
(C) 10x
(D) **8x + 2**

3 Score 3 NC.8.EE.7

Solve: `4(x - 2) = 20`

Answer: 7

1. $4(x - 2) = 20$ ↓ Distribute 4
2. $4x - 8 = 20$; $+8 \quad +8$ ↓ Add 8
3. $\frac{4x}{4} = \frac{28}{4}$ ↓ Divide by 4
4. $x = 7$

3.3 Variables on both sides NC.8.EE.7

When x is on **both sides**, move the x-terms to one side first (subtract the smaller x-term from both sides). Then solve the two-step equation that is left.

Like terms

5x and 3x are like terms, so 5x − 3x = 2x

Both sides

the left of the = sign and the right of the = sign

The 5-step plan:

1. Distribute (if there are parentheses).
2. Combine like terms on each side.
3. Move the x-terms to one side.
4. Move the numbers to the other side.
5. Divide to get x alone.

Watch out: Subtract the x-term from **both** sides, not just the side it is on.

1. $5x + 2 = 3x + 10$; $-3x \quad -3x$ ↓ Subtract 3x
2. $2x + 2 = 10$; $-2 \quad -2$ ↓ Subtract 2
3. $\frac{2x}{2} = \frac{8}{2}$ ↓ Divide by 2
4. $x = 4$

Walk through: `Solve 7x + 3 = 4x + 15`

1. x is on both sides. What move puts the x's together?
   Subtract 4x from both sides → `3x + 3 = 15`
2. Undo the + 3.
   Subtract 3 from both sides → `3x = 12`

3 Get x alone.
Divide both sides by 3 → x = 4

Check your understanding

1 Score 3 NC.8.EE.7

What is a good **first** move for $6x - 1 = 2x + 11$?

- (A) **Subtract 2x from both sides**
- (B) Divide both sides by 6
- (C) Add 2x to both sides
- (D) Subtract 6x from the right side only

2 Score 3 NC.8.EE.7

Solve: $6x - 1 = 2x + 11$

Answer: **3**

1. $6x - 1 = 2x + 11$ → Subtract 2x
   $-2x \qquad -2x$
2. $4x - 1 = 11$ → Add 1
   $+1 \qquad +1$
3. $\frac{4x}{4} = \frac{12}{4}$ → Divide by 4
4. $x = 3$

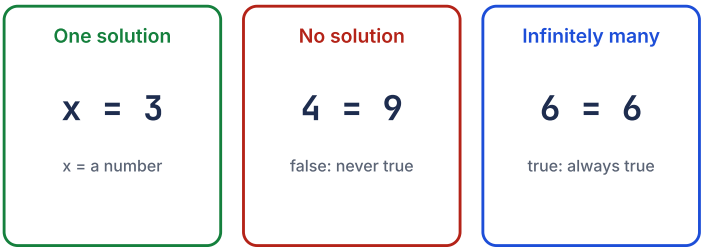
3.4 One, none, or infinitely many solutions NC.8.EE.7a

Solve as usual. If the x's cancel, look at what is left. **False** (like 4 = 9) means **no solution**. **True** (like 6 = 6) means **infinitely many solutions**. If you get x = a number, there is **one solution**.

No solution
no number makes the equation true

Infinitely many solutions
every number makes the equation true

One solution
exactly one number works, like x = 3



Look at the LAST line after the x's are gone.

`2x + 4 = 2x + 9 → 4 = 9 (false) → no solution`

`3(x + 2) = 3x + 6 → 6 = 6 (true) → infinitely many`

Watch out: If the x's cancel, you are not stuck. The last line tells you the answer.

Walk through: `How many solutions? 3(x + 2) = 3x + 6`

1. First move?
   Distribute the 3 → `3x + 6 = 3x + 6`
2. Subtract 3x from both sides. What is left?
   `6 = 6` → `6 = 6`
3. 6 = 6 is true. How many solutions?
   Infinitely many solutions → 6 = 6 is true → infinitely many

Check your understanding

1 Score 3 NC.8.EE.7a

How many solutions does **`2x + 4 = 2x + 9`** have?

- **(A) No solution**
- (B) One solution
- (C) Infinitely many solutions

2 Score 3 NC.8.EE.7a

A last line says **`5 = 5`**. How many solutions?

- **(A) Infinitely many solutions**
- (B) One solution
- (C) No solution

3 Score 3 NC.8.EE.7a

How many solutions does $4x + 1 = 2x + 7$ have?

A One solution

B No solution

C Infinitely many solutions

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1 $4x + 1 = 2x + 7$

$-2x \quad -2x$

↓ Subtract 2x

2 $2x + 1 = 7$

$-1 \quad -1$

↓ Subtract 1

3 $\frac{2x}{2} = \frac{6}{2}$

↓ Divide by 2

4 $x = 3$

3.5 Inequalities: solve and graph NC.8.EE.7

Solve an inequality like an equation, with **one special rule**: when you **multiply or divide by a negative number, flip the sign**. Then graph it on a number line.

< and >

less than, greater than: open circle (the number is not included)

≤ and ≥

at most, at least: closed circle (the number is included)

Flip

turn the sign around: < becomes >, ≤ becomes ≥



x > 2: open circle at 2, shade right

Watch out: Only flip when you **multiply or divide by a negative**. Adding or subtracting never flips.

$-2x < 6$

$\div(-2)\ \div(-2)$

$x > -3$



Walk through: Solve 2x + 1 ≥ 7 and graph it

1. Undo the + 1.
   Subtract 1 from both sides: 2x ≥ 6 → 2x ≥ 6
2. Undo the × 2. Do we flip?
   Divide by 2. No flip: x ≥ 3 → x ≥ 3
3. Open or closed circle at 3?
   Closed: ≥ includes 3 → closed circle at 3
4. Which way do we shade?
   Right: numbers greater than 3 → shade to the right

Check your understanding

1 Score 3 NC.8.EE.7

Solve: $x + 4 > 9$

(A) x ≥ 5
(B) **x > 5**
(C) x < 5
(D) x > 13

2 Score 3 NC.8.EE.7

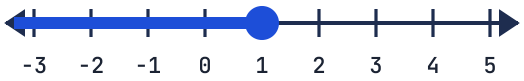
Solve: $-2x < 8$

(A) **x > −4**
(B) x < −4
(C) x < −16
(D) x > 4

3 Score 3 NC.8.EE.7

Which graph shows $x \leq 1$?

A



B



C

